Bandits for Online Exploration in Deep Recommenders

Third Year Project: Project Management Report

Sulay Chaudhry (5567816)
Department of Computer Science
University of Warwick

Supervised by Prof. Florin Ciucu

Third Year, 2025 – 2026

WARWICK
THE UNIVERSITY OF WARWICK

Introduction

This report outlines how this project was organised and managed throughout the year, with consideration of the difficulties encountered and overcome in completing this work. Given the research demands of this project in bridging often-disjoint elements of the recommender literature, this extended far beyond pure software development considerations.

Development methodology

An iterative methodology was adopted for this project from the beginning, aware of the importance of continual research and development as ideas from papers may look promising on the surface, but fail to generalise or perform exactly as described when implemented. This was a major risk given that there were many possible failure modes: an approach may generalise poorly to KuaiRec [1], the dataset used in this work; a two-tower approach may have unexpected impact on the downstream bandits; the bandits may interact unexpectedly with upstream two-tower embedding quality; and so on.

At the start, this took the shape of rapidly iterating with two-tower and the bandits on KuaiRec to gauge the effects; later, as intuition was built for the wider recommender literature and for the nature of the results, this meant that larger goals such as working on the simulator or the CLI became the whole focus of certain development sprints.

Time management and organisation

Time management for the project involved two layers of organisation: both a project-driven timeline, on the larger scale, and directed weekly/daily planning. The project timeline was originally formalised when writing the specification as a set of 2-week sprints each with a predefined purpose, seen in Figure 1; this was later found to be inadequate, as it underestimated the significant time that would be spent on research, and due to heavy coursework deadlines in Term 1. This was corrected during the writing of the Progress Report through a new Gantt chart timeline (Figure 2) for the rest of the project.

Period	Week	Sprint focus
Term 1	1 2	**Project Specification**
	3 4	Literature review
	5 6	Iterative prototyping of deep-learning & bandit subcomponents
	7 8	Iterative prototyping of combined hybrid system
	9 10	Initial development of test suite
Christmas Holiday	All	**Progress Report** & flexible buffer period
Term 2	1 2	Refinement of hybrid system design
	3 4	Refinement of testing suite
	5 6	Refinement of hybrid system in response to tests
	7 8	Completion of final testing suite
	9 10	**Final Report** & **Project Management Report**
Easter Holiday	All	**Final Report** & **Project Management Report**

Figure 1: Original project timeline from the Project Specification

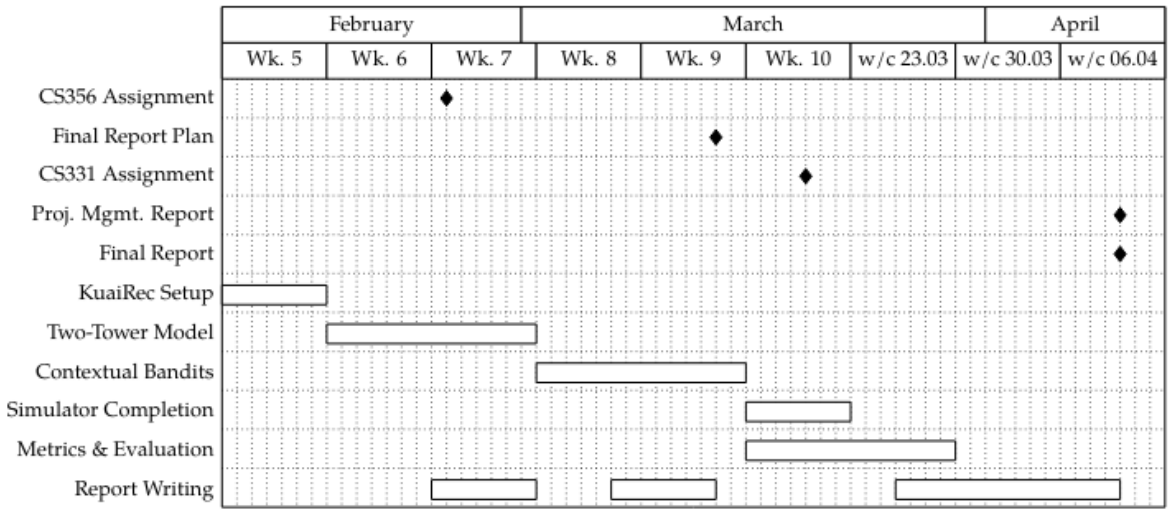

Figure 2: Project timetable (Gantt chart) from the Progress Report

Weekly progress was measured relative to this timeline and it did indeed inform the correct order of remaining work. Submission flexibility and support was provided by the department additional to this, making completion of the work under untimely difficulties in personal circumstances possible. As research and development was so iterative, the additional organisation took the form of weekly and daily checklists kept track of alongside ideas and research in a large multi-tab Google Doc (Figure 3).

Completing the two-tower system
- [x] ~~Training & validation loops~~
  - [x] ~~Training with cross-entropy loss, set up batch-size as something I can change, use weight decay to prevent overfitting, set up uniform negative sampling~~
    - [x] ~~Fix up embedding widths for user categorical features~~
    - (Masking is optional, can be added if it is seen to improve performance massively)
    - (Potentially add weight decay on optimiser)
  - [x] ~~Validation~~
    - [x] ~~Monitor batch-wise validation loss (trivial)~~
    - [x] ~~Monitor epoch-wise Recall@K (ensure masked of course to properly be on the validation set)~~
    - [x] ~~Monitor some other validation metric as is useful to confirm that the two-tower is working (or that embedding quality is good)~~
      - [x] ~~NDCG@K complete~~
- [x] ~~Visualising these metrics in some way by plotting them; makes things a lot easier~~
  - [x] ~~Visualising pre-NDCG@K~~
  - [x] ~~Visualising NDCG@K~~
  - (Can fix up how the plotting looks; the Recall graph has too much text)
- [x] ~~Freezing this model successfully~~
  - [x] ~~Picking what I stop my model on: when is it optimal? (Recall@50 chosen)~~
  - (It should also be saved with information about what hyperparameters it had etc. -> this means I can actually conclusively say what was happening)

Figure 3: A daily checklist used while working on the two-tower system

Regular supervisor meetings with Florin (Figure 4) were also essential to gauging progress and gaining his insight on the research work explored that week. Initially, it was difficult to condense and convey the scope of all of the research – especially given Florin's limited time – however over time it became easier to frame the concepts and discuss the progress, especially after directly showing him the codebase and plotted bandit results. Florin was also incredibly supportive through the personal circumstances, especially near the end of the project, for which transparency was important both over Teams and in person.

14:30 - 15:00, Tue 17th Mar 2026 | Results Discussion | Face-to-face meeting | Approved

Showed Florin all of my result plots, discussed the seen benefits of the contextual bandits, noted limitations of the linear model. Briefly mentioned misspecification due to the reward environment remaining nonlinear relative to the quality of the two-tower embeddings; Florin noted that additional rigour is needed to defend the role of Thompson Sampling hyperparameters, discussed their presence in contextual bandit papers. Agreed to meet again on Friday the 20th.

Face-to-face meeting between Sulay Chaudhry and Florin Ciucu. Created by Sulay Chaudhry, 16:28 Tue 17th March 2026. Approved by Florin Ciucu, 09:43 Mon 23rd March 2026

Figure 4: A meeting record with Florin, logged on Tabula

Risks and unforeseen issues

Risks on the research side extended those already discussed: largely that of the disconnect between the literature and what would actually work within this context, only discoverable directly through implementation. This was mitigated further by intentionally making a point of smoke tests that could be done incredibly quickly and ruled out ineffective lines of research or literature misunderstandings quickly, without necessarily worrying yet about the larger work of adding an additional evaluative axis rigorous enough for formally presenting.

Git was used throughout and the project was pushed frequently to GitHub to ensure progress would not be lost. This also made it easier to work across both a laptop and a desktop, the latter of which was used for its GPU when iterating. Later work shifted to the use of the Batch Compute System for more compute, and reduced the reliance on access to the desktop for progress to be made.

Another major risk was that of scope creep, and aiming for an unnecessarily high bar for project success – this was indeed the case in the original specification, which aimed to juggle not only exploit-explore concerns but also a holistic look at scalability; the latter was descoped in this work during the Progress Report due to existing work demonstrating scalability for bandits [2, 3]. This meant that the project could avoid aspects such as designing clustering or graph algorithms that would have significantly increased development time, while potentially hurting result generalisability.

A challenge that was not foreseen, however, was that of methodological inadequacy and inconsistency within the recommender literature itself, for both supervised approaches and the bandits. It was difficult to gauge when certain ideas weren't working because of an assumed misunderstanding and unfamiliarity with the wider literature, and when certain papers were, in hindsight, likely yet another example of poorly reproducible and cherry-picked results identified by Ferrari Dacrema et al. [4].

This was a central reason behind research taking significantly longer than expected, as it was always crucial to trace back the implications of claims highlighted in beyond-accuracy and debiasing papers. It was also not foreseen that papers would often neglect a discussion of the assumptions underlying their paradigm or datasets. This was a significant challenge for contextualising the hybrid system, as the supervised and bandit paradigms were found to use fundamentally different evaluation procedures whose alignment come under scrutiny in this work.

Spending too long tempted by further analysis, without formalising all the results first, was also a risk; it was crucial to recentre back to the main deliverables seen in Figure 2 to ensure project completion. Discussion with Florin was also helpful in this regard, especially about the methodological inconsistencies in the wider literature.

Legal, social, ethical and professional issues

No legal, social, ethical or professional issues were identified with relation to this work. Proper attribution to KuaiRec has been made, and the dataset has not been redistributed in any manner, for which the CC BY-SA clause would apply. It is notable that anonymised features from KuaiRec are used for this work; the raw dataset was not used for which deanonymisation is a more significant concern.

References

[1] Chongming Gao et al. 'KuaiRec: A fully-observed dataset and insights for evaluating recommender systems'. In: *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. 2022, pp. 540–550.

[2] Yu Song et al. 'Show me the whole world: Towards entire item space exploration for interactive personalized recommendations'. In: *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining*. 2022, pp. 947–956.

[3] Xinyang Yi et al. 'Online matching: A real-time bandit system for large-scale recommendations'. In: *Proceedings of the 17th ACM Conference on Recommender Systems*. 2023, pp. 403–414.

[4] Maurizio Ferrari Dacrema, Paolo Cremonesi and Dietmar Jannach. 'Are we really making much progress? A worrying analysis of recent neural recommendation approaches'. In: *Proceedings of the 13th ACM conference on recommender systems*. 2019, pp. 101–109.